Linear Algebra and Its Applications

Section 4.1: Vector Spaces and Subspaces

Detailed worked solutions

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Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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4.1 Vector spaces and subspaces

A vector need not be an arrow: a polynomial, a matrix, or a function can be a vector when addition and scalar multiplication satisfy the vector-space axioms. A subset H of a known vector space is a subspace when (i) it contains the zero vector, (ii) $u + v \in H$ for *every* $u, v \in H$, and (iii) $cu \in H$ for *every* $u \in H$ and real scalar c. One counterexample disproves closure; one successful example does not prove it. A span is automatically a subspace.

Practice problem 4.1.P1: An affine line fails scalar closure

What we must show. For $H = \{(3s, 2 + 5s) : s \in \mathbb{R}\}$, find a vector whose scalar multiple leaves H. Take $s = 0$. Then $u = (0, 2) \in H$. Its multiple $2u = (0, 4)$ is not in H: the equation $3s = 0$ would require $s = 0$, but then the second coordinate would be 2, not 4. Thus scalar multiplication is not closed on H.

Conclusion. H is not a vector space under the usual operations.

Practice problem 4.1.P2: Every generator belongs to its span

For $W = \text{Span}\{v_1, \ldots, v_p\}$ and a fixed k between 1 and p, write

$$v_k = 0v_1 + \cdots + 0v_{k-1} + 1v_k + 0v_{k+1} + \cdots + 0v_p.$$

This is a permitted linear combination of the generators, hence $v_k \in W$. In particular, choose $k = 1$ for the first part of the question. A span contains all its generators, even if some are zero or repeated.

Exercise 4.1.1: The first quadrant

(a) Addition preserves both inequalities.

If $u = (u_1, u_2)$ and $v = (v_1, v_2)$ have nonnegative coordinates, then $u_1 + v_1 \geq 0$ and $u_2 + v_2 \geq 0$. Therefore $u + v$ is still in the first quadrant.

(b) Negative scalars reveal the obstruction.

The vector $u = (1, 0)$ belongs to the set, but $(-1)u = (-1, 0)$ does not.

Conclusion. Addition is closed, but scalar multiplication is not; the first quadrant is not a vector space.

Exercise 4.1.2: The first and third quadrants together

(a) Any scalar multiple stays in the set.

Here $W = \{(x, y) : xy \geq 0\}$. If $u = (x, y) \in W$, then the product of the coordinates of cu is $(cx)(cy) = c^2xy \geq 0$, since $c^2 \geq 0$. This proves closure under multiplication by every real scalar, including zero.

(b) Addition can leave the set.

Choose $u = (2, 1)$ and $v = (-1, -2)$. Both coordinate products equal $2 > 0$, so both vectors are in W. Their sum is $(1, -1)$, whose coordinate product is $-1 < 0$. Thus $u + v \notin W$.

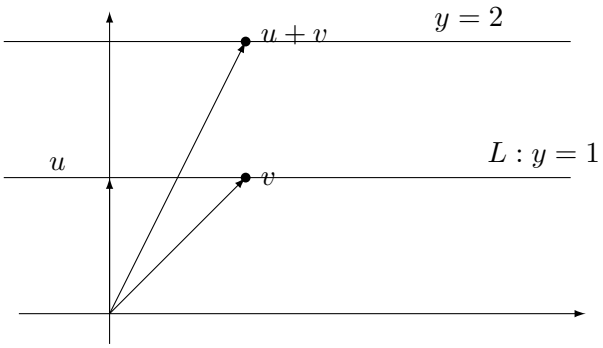
Conclusion. W passes the scalar test but fails the addition test.

Exercise 4.1.3: The closed unit disk

The vector $u = (1, 0)$ is in the disk because $1^2 + 0^2 = 1$. But $2u = (2, 0)$ is outside it, since $2^2 + 0^2 = 4 > 1$. Thus the disk is not closed under scalar multiplication and is not a subspace. It is not enough that the disk contains zero: closure must hold for arbitrarily large as well as small scalars.

Exercise 4.1.4: A translated line is not closed under addition

What we must show. Supply a geometric counterexample, rather than just assert that a line misses zero. Consider the line $L = \{(x, 1) : x \in \mathbb{R}\}$. Take $u = (0, 1)$ and $v = (1, 1)$. Both endpoints lie on L, but $u + v = (1, 2)$ lies on the parallel line $y = 2$. The parallelogram construction for adding u and v therefore ends off L.



More generally write any line not through zero as $p + \text{Span}\{d\}$, where $p \notin \text{Span}\{d\}$. Then $p \in L$ but $2p \notin L$, for otherwise $2p = p + td$ would imply $p = td$, a contradiction. Adding p to itself already fails.

Exercise 4.1.5: Multiples of a single polynomial

The set $\{at^2 : a \in \mathbb{R}\}$ equals $\text{Span}\{t^2\}$, a subspace of P_2 (and of P_n for every $n \geq 2$). Explicitly its zero is obtained with $a = 0$, $at^2 + bt^2 = (a + b)t^2$ stays in the set, and $c(at^2) = (ca)t^2$ stays in the set.

Conclusion. It is a subspace.

Exercise 4.1.6: A fixed quadratic coefficient

The set consists of polynomials $a + t^2$, whose coefficient of t^2 is always 1. It does not contain the zero polynomial: no choice of a makes $a + t^2$ zero for every t. Equivalently multiplying t^2 by zero produces a polynomial outside the set. Thus this subset of P_2 is not a subspace.

Exercise 4.1.7: Integer coefficients and real scalars

The constant polynomial $p(t) = 1$ has integer coefficients and degree at most 3. The real scalar $1/2$ is permitted in the ambient vector space, but $(1/2)p(t) = 1/2$ does not have integer coefficients. Scalar closure fails.

Conclusion. This set is not a real subspace of P_3. The choice of scalar field matters.

Exercise 4.1.8: Polynomials vanishing at zero

Let $H = \{p \in P_n : p(0) = 0\}$. The zero polynomial is in H. If $p, q \in H$, then $(p+q)(0) = p(0)+q(0) = 0$. For real c, $(cp)(0) = cp(0) = 0$. These equations prove both closure properties; sums and scalar multiples still have degree at most n.

Conclusion. H is a subspace. For $n \geq 1$ it is $\text{Span}\{t, t^2, \ldots, t^n\}$; for $n = 0$ it is just the zero polynomial.

Exercise 4.1.9: A one-parameter line in three dimensions

Factor out the arbitrary real parameter:

$$(-2t, 5t, 3t) = t(-2, 5, 3).$$

Thus choose $v = (-2, 5, 3)$, and $H = \text{Span}\{v\}$. This is equality in both directions: every displayed vector is a multiple of v, and every real multiple of v is obtained by choosing that number as t. A span is a subspace, so H is a subspace of $\mathbb{R}^3$.

Exercise 4.1.10: A line with a zero middle coordinate

Here $(3t, 0, -7t) = t(3, 0, -7)$. As t runs through all real numbers, these are exactly all linear combinations of the single vector $(3, 0, -7)$. Consequently $H = \text{Span}\{(3, 0, -7)\}$ is a subspace of $\mathbb{R}^3$. The zero middle coordinate does not make it a different kind of vector space.

Exercise 4.1.11: Separate two free parameters

For arbitrary $b, c \in \mathbb{R}$,

$$\begin{pmatrix} 2b + 3c \\ -b \\ 2c \end{pmatrix} = b \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} + c \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix}.$$

Take $u = (2, -1, 0)$ and $v = (3, 0, 2)$. Every original vector is in their span, and an arbitrary linear combination $bu + cv$ has exactly the required form. Therefore $W = \text{Span}\{u, v\}$, and spans are subspaces of the ambient vector space.

Exercise 4.1.12: Separate two parameters in four dimensions

Collect the coefficients of s and t in each coordinate:

$$\begin{pmatrix} 2s+4t \\ 2s \\ 2s-3t \\ 5t \end{pmatrix} = s\begin{pmatrix} 2 \\ 2 \\ 2 \\ 0 \end{pmatrix} + t\begin{pmatrix} 4 \\ 0 \\ -3 \\ 5 \end{pmatrix}.$$

Since s, t are unrestricted, W is precisely the span of the two displayed constant vectors. The span theorem proves that W is a subspace of $\mathbb{R}^4$.

Exercise 4.1.13: A set of generators versus its span

What we must show. Distinguish membership in a finite set from membership in all its linear combinations. The given vectors are $v_1 = (1, 0, -1)$, $v_2 = (2, 1, 3)$, $v_3 = (4, 2, 6)$, and $w = (3, 1, 2)$.

(a) Inspect the three individual vectors.

The vector w equals none of v_1, v_2, v_3, so $w \notin \{v_1, v_2, v_3\}$. Those three vectors are distinct, so that set has exactly three elements.

(b) Count the span.

Since $v_1 \neq 0$, the multiples tv_1 give distinct vectors for distinct real t. Thus the span contains infinitely many vectors.

(c) Find an actual combination.

Adding coordinates gives $v_1 + v_2 = (1+2, 0+1, -1+3) = (3, 1, 2) = w$. Hence w belongs to the span even though it is not one of its three generators.

Exercise 4.1.14: Another vector in the same plane

Use the generators from the preceding exercise and $w = (1, 3, 14)$. Because $v_3 = 2v_2$, combinations of the first two vectors suffice. Set $av_1 + bv_2 = w$. Coordinate comparison gives

$$a + 2b = 1, \qquad b = 3, \qquad -a + 3b = 14.$$

The middle equation gives $b = 3$, the first gives $a = -5$, and the last checks: $-(-5) + 3(3) = 14$. Therefore $w = -5v_1 + 3v_2$ lies in the required span.

Exercise 4.1.15: A coordinate forced to be negative one

Every vector in the set has second coordinate -1. The zero vector has second coordinate zero and is therefore absent. A vector space must contain zero, so no spanning set for this W exists: W is not a vector space.

Exercise 4.1.16: A coordinate forced to be one

Every vector has first coordinate 1, regardless of the values of a, b. Thus zero is absent. This violates the first subspace condition and shows that W is not a vector space under the inherited operations.

Exercise 4.1.17: Three parameters, each with its own coefficient vector

Expanding coordinate by coordinate gives

$$\begin{pmatrix} 2a-b \\ 3b-c \\ 3c-a \\ 3b \end{pmatrix} = a\begin{pmatrix} 2 \\ 0 \\ -1 \\ 0 \end{pmatrix} + b\begin{pmatrix} -1 \\ 3 \\ 0 \\ 3 \end{pmatrix} + c\begin{pmatrix} 0 \\ -1 \\ 3 \\ 0 \end{pmatrix}.$$

With all three parameters arbitrary, W equals the span of these three vectors, so it is a subspace of $\mathbb{R}^4$. These vectors form a valid spanning set; one need not prove independence merely to supply a spanning set.

Exercise 4.1.18: A permanently zero coordinate is harmless

The required decomposition is

$$\begin{pmatrix} 4a+3b \\ 0 \\ a+3b+c \\ 3b-2c \end{pmatrix} = a\begin{pmatrix} 4 \\ 0 \\ 1 \\ 0 \end{pmatrix} + b\begin{pmatrix} 3 \\ 0 \\ 3 \\ 3 \end{pmatrix} + c\begin{pmatrix} 0 \\ 0 \\ 1 \\ -2 \end{pmatrix}.$$

Every such combination has the specified form and vice versa. Thus the three coefficient vectors span W, which is a subspace. A coordinate fixed at zero is compatible with subspaces; a coordinate fixed at nonzero 1 was not.

Exercise 4.1.19: The space of oscillating functions

Fix ω and let $f(t) = \cos(\omega t)$ and $g(t) = \sin(\omega t)$. The set of functions $c_1 f + c_2 g$, with arbitrary real c_1, c_2, is exactly $\mathrm{Span}\{f, g\}$ inside the vector space of all real-valued functions on $\mathbb{R}$. Hence it is a subspace. Explicitly, adding two functions adds their two coefficients, and scalar multiplication multiplies both coefficients; neither operation changes the fixed frequency ω. Choosing both coefficients zero gives the zero function. The span argument remains valid if $\omega = 0$.

Exercise 4.1.20: Continuous functions and an endpoint condition

(a) State the three required facts.

The function identically zero on $[a, b]$ is continuous. The sum of any two continuous functions on $[a, b]$ is continuous, and every real scalar multiple of a continuous function there is continuous. These are the subspace tests, so $C[a, b]$ is a subspace of all real-valued functions on that interval.

(b) Preserve equality of the endpoint values.

Let $H = \{f \in C[a,b] : f(a) = f(b)\}$. The zero function is in H. If $f, g \in H$, then

$$(f+g)(a) = f(a) + g(a) = f(b) + g(b) = (f+g)(b).$$

For every real c, $(cf)(a) = cf(a) = cf(b) = (cf)(b)$. Both new functions are continuous by part (a), so both belong to H.

Conclusion. H is a subspace of $C[a,b]$.

Exercise 4.1.21: Upper triangular matrices form a subspace

Take $H = \{\left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right) : a, b, d \in \mathbb{R}\}$. Zero is obtained with $a = b = d = 0$. Adding two such matrices still gives a zero entry in position $(2,1)$, and multiplying by a scalar preserves that zero entry. Therefore H is a subspace of $M_{2\times 2}$. One explicit spanning set is

$$\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}.$$

Exercise 4.1.22: A homogeneous matrix equation

Let F be the fixed 3×2 matrix, and let $H = \{A \in M_{2\times 4} : FA = 0\}$. The zero 2×4 matrix is in H, because multiplying it by F gives the zero 3×4 matrix. For $A, B \in H$, distributivity gives $F(A+B) = FA + FB = 0$. For real c, $F(cA) = c(FA) = 0$. Thus both $A + B$ and cA belong to H.

Conclusion. H is a subspace of $M_{2\times 4}$. The matrix sizes ensure each product is defined.

Exercise 4.1.23: Five statements about vectors and subspaces

(a) **False.** A zero function must vanish at *every* input. For example $f(t) = t$ has $f(0) = 0$ but is not the zero function.

(b) **False.** Vectors can be functions, matrices, or polynomials, not only arrows in three-dimensional space. An element of P_2, such as t^2, is a vector in that vector space.

(c) **False.** Containing zero is necessary but insufficient. The unit disk contains zero but fails scalar closure, as Exercise 4.1.3 shows.

(d) **True.** A subspace inherits the vector-space operations and all axioms; the subspace conditions ensure the operations stay inside it.

(e) **False.** The chapter introduction describes the major shuttle control systems in terms of *digital*, not analog, signals.

Exercise 4.1.24: Five more statements, including the closure quantifiers

(a) **True.** A vector means an element of a vector space; its physical appearance is irrelevant.

(b) **True.** The identity $(-1)u = -u$ follows from distributivity and uniqueness of the additive inverse; Exercise 4.1.29 supplies the proof.

(c) **True.** Every vector space is a subspace of itself, using its own zero vector and the two operations.

(d) **False.** As defined here, elements of $\mathbb{R}^2$ are ordered pairs, not ordered triples in $\mathbb{R}^3$. The plane $\{(x, y, 0) : x, y \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^3$ isomorphic to $\mathbb{R}^2$, but it is a different set.

(e) **False as written.** The stated conditions merely say that $u, v, u + v$ are in H and that cu is in H for particular choices. Closure requires these conclusions for *all* $u, v \in H$ and *all* real c. For instance $H = \{(0, 0), (1, 0)\}$ satisfies the listed conditions with $u = v = 0$ and $c = 0$ but is not closed under multiplication by 2. If the omitted universal quantifiers are supplied, the resulting statement is exactly the correct subspace test.

Exercise 4.1.25: Uniqueness of the zero vector

The missing axiom is **4**, the additive-identity axiom (with the commutative form justified by Axiom 2, as the exercise specifies). If w is another additive identity, then using it as an identity gives $0 + w = 0$, whereas using the original zero as an identity gives $0 + w = w$. The same expression equals both 0 and w, hence $w = 0$.

Exercise 4.1.26: Uniqueness of the additive inverse

The blanks are **(a) 3, (b) 5, (c) 4**. To see each step, suppose $u + w = 0$ and add $-u$ to both sides:

$$(-u) + (u + w) = (-u) + 0.$$

Associativity (Axiom 3) changes the left side to $((-u) + u) + w$. The inverse axiom (Axiom 5, with commutativity) changes $(-u)+u$ to zero. The identity axiom (Axiom 4) then changes $0+w = (-u)+0$ to $w = -u$.

Conclusion. There can be only one additive inverse of a given vector.

Exercise 4.1.27: Multiplication by the scalar zero

The blanks are **(a) 8, (b) 3, (c) 5, (d) 4**. Distributivity over scalar addition (Axiom 8) gives $0u = (0 + 0)u = 0u + 0u$. Add the additive inverse $-(0u)$ to both sides. Associativity (3) groups the last two terms on the right as $0u + (-(0u))$; the inverse axiom (5) changes each such pair to zero. The result is $0 = 0u + 0$, which the identity axiom (4) reduces to $0 = 0u$. Here the first zero in $0u$ is a scalar; the zero on the right is a vector.

Exercise 4.1.28: A scalar times the zero vector

The blanks are **(a) 4, (b) 7, (c) 3, (d) 5, (e) 4**. First $0 = 0 + 0$ by the identity axiom (4). Thus $c0 = c(0 + 0) = c0 + c0$ by distributivity over vector addition (7). Add the inverse of $c0$ to both sides. Associativity (3) allows the right-hand inverse pair to be grouped; Axiom 5 reduces the inverse pairs to zero; and Axiom 4 removes the remaining added zero. Hence $0 = c0$. This holds for every scalar c, including negative and zero scalars.

Exercise 4.1.29: The scalar negative one gives the additive inverse

Axiom 10 says $u = 1u$. Axiom 8 then gives

$$u + (-1)u = 1u + (-1)u = (1 + (-1))u = 0u = 0,$$

where the last equality is Exercise 4.1.27. Therefore $(-1)u$ is an additive inverse of u. By uniqueness of that inverse (Exercise 4.1.26), $(-1)u = -u$.

Exercise 4.1.30: A nonzero scalar cannot annihilate a nonzero vector

Suppose $cu = 0$ and $c \neq 0$. Then $1/c$ is an allowed scalar. Multiply both sides by it and use Axioms 9 and 10 and Exercise 4.1.28:

$$u = 1u = ((1/c)c)u = (1/c)(cu) = (1/c)0 = 0.$$

The nonzero hypothesis is needed to form $1/c$; when $c = 0$, the equation $cu = 0$ holds for every u and does not force $u = 0$.

Exercise 4.1.31: The span is the smallest containing subspace

Suppose the subspace H contains u and v. An arbitrary member of $\text{Span}\{u, v\}$ has form $au + bv$ for real a, b. Scalar closure puts au, bv in H, and additive closure puts $au+bv$ in H. Thus $\text{Span}\{u, v\} \subseteq H$. Since the span itself is a subspace that contains u, v, it is the smallest such subspace, in the sense of inclusion.

Exercise 4.1.32: Intersection works; union can fail

Check the intersection's three properties.

Both H and K contain zero, so their intersection does too. If $u, v \in H \cap K$, then they belong to both subspaces; hence $u + v$ belongs to both, so $u + v \in H \cap K$. Likewise any scalar multiple cu belongs to both, hence to their intersection. Thus $H \cap K$ is a subspace.

Give a concrete counterexample for a union.

Let $H = \{(x, 0) : x \in \mathbb{R}\}$ and $K = \{(0, y) : y \in \mathbb{R}\}$, the two coordinate axes. Their union contains $(1, 0)$ and $(0, 1)$, but not their sum $(1, 1)$. Thus a union of subspaces need not be a subspace.

Exercise 4.1.33: The sum of two subspaces

(a) Verify closure, keeping each summand in its own subspace.

Zero is $0 + 0$ with the first zero in H and the second in K. For $x = h_1 + k_1$ and $y = h_2 + k_2$ in $H + K$, associativity and commutativity give

$$x + y = (h_1 + h_2) + (k_1 + k_2) \in H + K,$$

since $h_1 + h_2 \in H$ and $k_1 + k_2 \in K$. For a real scalar c, $cx = ch_1 + ck_1 \in H + K$ by both scalar closures. Therefore $H + K$ is a subspace of V.

(b) Put the missing summand equal to zero.

If $h \in H$, then $h = h + 0$ with $0 \in K$, so $H \subseteq H + K$. If $k \in K$, write $k = 0 + k$ to obtain $K \subseteq H + K$. With their inherited operations, both are subspaces of $H + K$.

Exercise 4.1.34: Spanning the sum by combining the generators

Let $H = \text{Span}\{u_1, \ldots, u_p\}$ and $K = \text{Span}\{v_1, \ldots, v_q\}$.

Every element of the sum belongs to the combined span.

If $x = h + k \in H + K$, write $h = \sum_i a_i u_i$ and $k = \sum_j b_j v_j$. Then $x = \sum_i a_i u_i + \sum_j b_j v_j$, a combination of the combined list.

Every element of the combined span belongs to the sum.

Conversely split any combination of that combined list into its u-part $h \in H$ and its v-part $k \in K$. It then equals $h + k \in H + K$. These two inclusions prove equality of the sets.

Exercise 4.1.35: A checked representation in four dimensions

What we must show. Express $w = (9, -4, -4, 7)$ as a combination of the given v_1, v_2, v_3. Set $av_1 + bv_2 + cv_3 = w$. The augmented system and its reduced form are

$$\left[\begin{array}{rrr|r} 8 & -4 & -7 & 9 \\ -4 & 3 & 6 & -4 \\ -3 & -2 & -5 & -4 \\ 9 & -8 & -18 & 7 \end{array}\right] \sim \left[\begin{array}{ccc|r} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

The three coefficient values are $1, -2, 1$. Rather than trust the reduction alone, check them in the original vectors:

$$\begin{pmatrix} 8 \\ -4 \\ -3 \\ 9 \end{pmatrix} - 2\begin{pmatrix} -4 \\ 3 \\ -2 \\ -8 \end{pmatrix} + \begin{pmatrix} -7 \\ 6 \\ -5 \\ -18 \end{pmatrix} = \begin{pmatrix} 8 + 8 - 7 \\ -4 - 6 + 6 \\ -3 + 4 - 5 \\ 9 + 16 - 18 \end{pmatrix} = \begin{pmatrix} 9 \\ -4 \\ -4 \\ 7 \end{pmatrix}.$$

Conclusion. $w = v_1 - 2v_2 + v_3$ belongs to the stated span.

Exercise 4.1.36: Test membership in a column span

The columns of A span precisely the vectors Ax. Therefore solve $Ax = y$:

$$\left[\begin{array}{rrr|r} 3 & -5 & -9 & -4 \\ 8 & 7 & -6 & -8 \\ -5 & -8 & 3 & 6 \\ 2 & -2 & -9 & -5 \end{array}\right] \sim \left[\begin{array}{ccc|r} 1 & 0 & 0 & -1/5 \\ 0 & 1 & 0 & -2/5 \\ 0 & 0 & 1 & 3/5 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

There is no inconsistent row. The candidate is $x = (-1/5, -2/5, 3/5)$. Checking the original multiplication gives

$$Ax = \tfrac{1}{5}(-3 + 10 - 27,\ -8 - 14 - 18,\ 5 + 16 + 9,\ -2 + 4 - 27) = (-4, -8, 6, -5) = y.$$

Conclusion. y is in the subspace spanned by the columns of A.

Exercise 4.1.37: Recognizing two multiple-angle functions

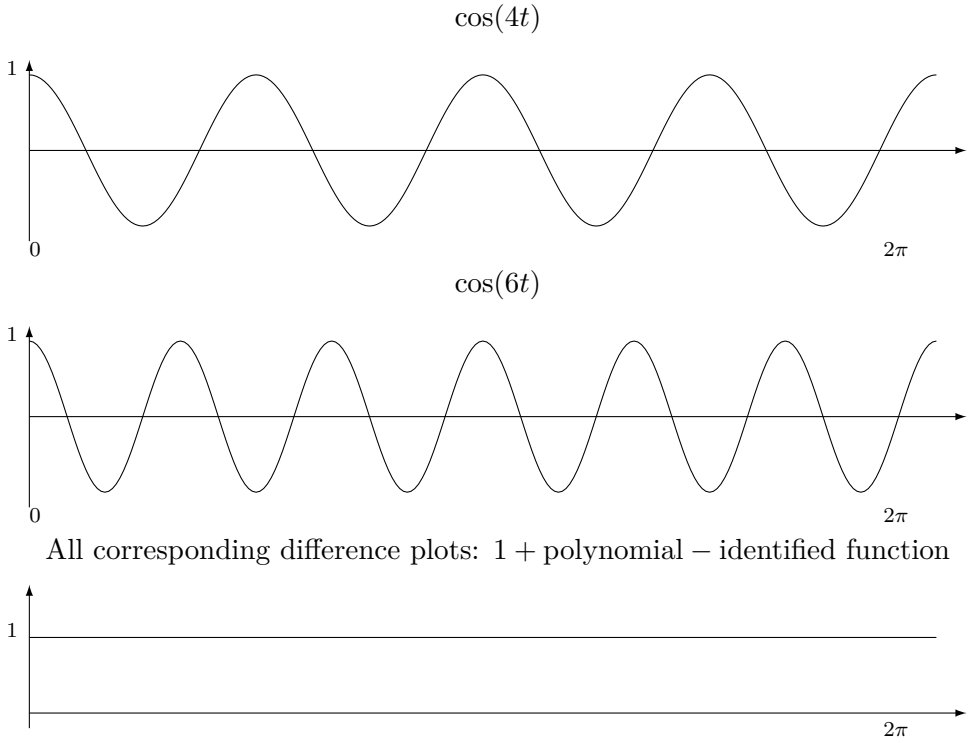
What we must show. Identify the two given polynomials in $\cos t$, then verify them algebraically and graphically. Write $c = \cos t$. The double-angle formula gives $\cos 2t = 2c^2 - 1$. Apply it a second time:

$$\cos 4t = 2(2c^2 - 1)^2 - 1 = 8c^4 - 8c^2 + 1 = f(t).$$

For the other polynomial use $\cos 3u = 4\cos^3 u - 3\cos u$ with $u = 2t$:

$$\cos 6t = 4(2c^2 - 1)^3 - 3(2c^2 - 1) = 32c^6 - 48c^4 + 18c^2 - 1 = g(t).$$

Thus the first graph has four full cosine oscillations on $[0, 2\pi]$ and the second has six. The requested difference graphs are exactly $1 + f(t) - \cos 4t = 1$ and $1 + g(t) - \cos 6t = 1$. A numerical graph supports the recognition, while the displayed identities prove it for every t.



Exercise 4.1.38: Recognizing three polynomials in sine

Put $s = \sin t$. The imaginary part of $(\cos t + i \sin t)^3 = \cos 3t + i \sin 3t$ gives

$$\sin 3t = 3s\cos^2 t - s^3 = 3s(1 - s^2) - s^3 = 3s - 4s^3 = f(t).$$

The double-angle identity $\cos 2t = 1 - 2s^2$ gives

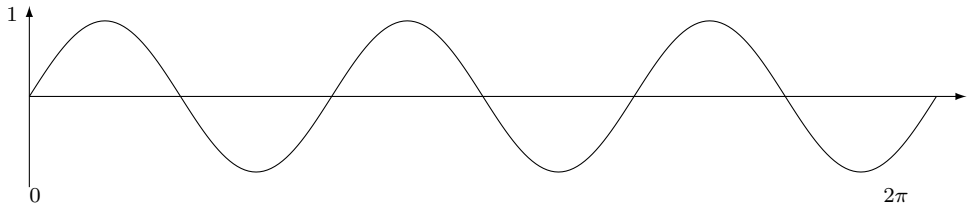
$$\cos 4t = 2(1 - 2s^2)^2 - 1 = 1 - 8s^2 + 8s^4 = g(t).$$

Likewise the imaginary part of the fifth power gives

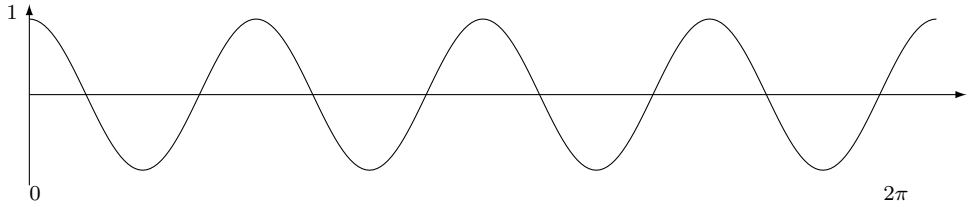
$$\begin{aligned}\sin 5t &= 5\cos^4 t\, s - 10\cos^2 t\, s^3 + s^5 \\ &= 5(1 - s^2)^2 s - 10(1 - s^2)s^3 + s^5 = 5s - 20s^3 + 16s^5 = h(t).\end{aligned}$$

(The power identities used here also follow by repeated angle-addition formulas.) The plots have three sine cycles, four cosine cycles, and five sine cycles, respectively, on $[0, 2\pi]$. Subtracting each identified function from one plus its original polynomial gives the constant 1, as requested.

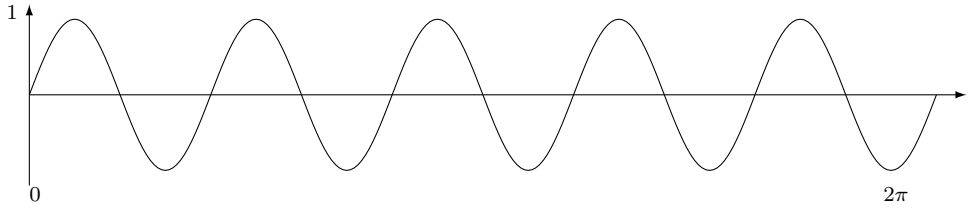
$\sin(3t)$



$\cos(4t)$



$\sin(5t)$



All corresponding difference plots: $1 +$ polynomial $-$ identified function

